

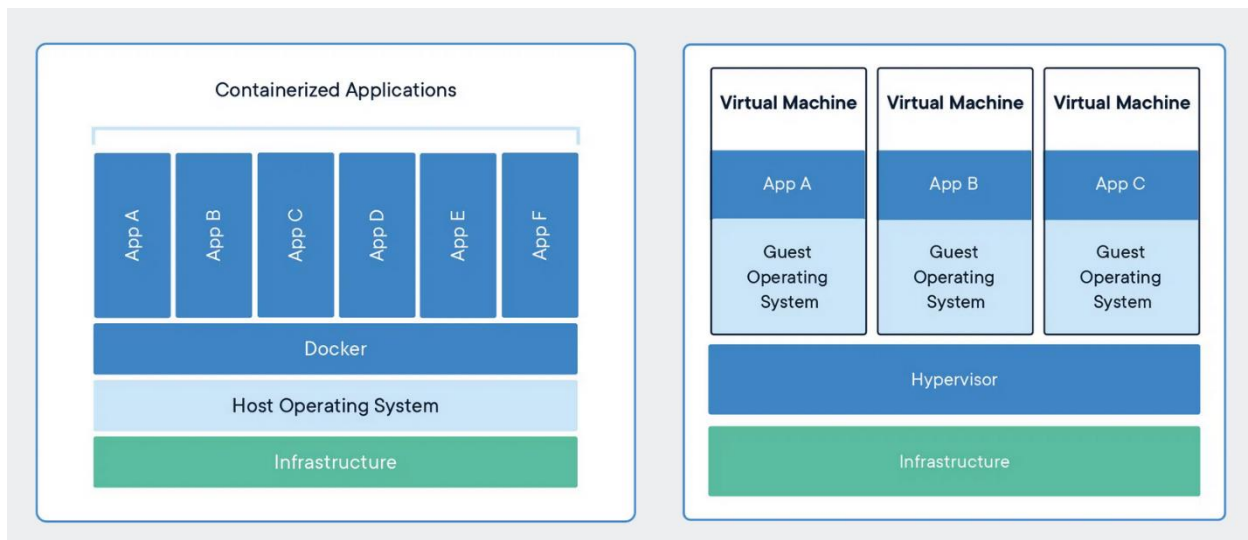
Migrating a web application into a Containers (AWS Ecs)

What is a container ?

A container is a standard unit of software that packages up code and all its dependencies so the application runs quickly and reliably from one computing environment to another. A Docker container image is a lightweight, standalone, executable package of software that includes everything needed to run an application: code, runtime, system tools, system libraries and settings.

What is Docker ?

Docker is a containerization platform that provides easy way to containerize your applications, which means, using Docker you can build container images, run the images to create containers and also push these containers to container registries such as DockerHub, Quay.io and so on.



What is Amazon Elastic Container Service?

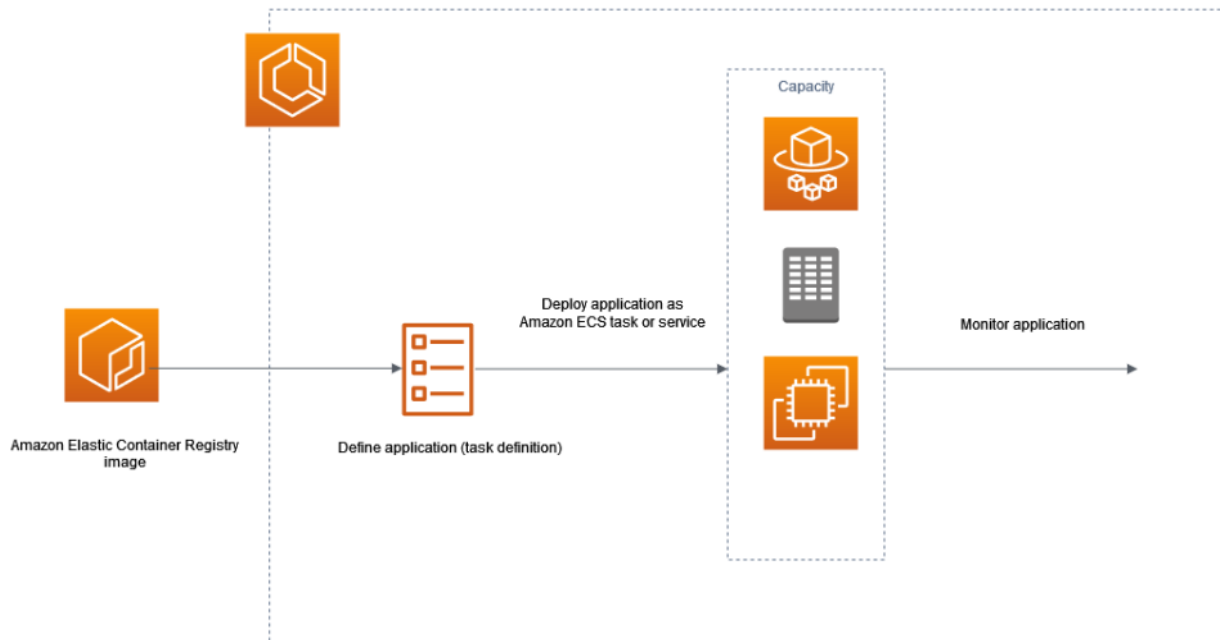
Amazon Elastic Container Service (Amazon ECS) is a fully managed container orchestration service that helps you easily deploy, manage, and scale containerized applications. As a fully managed service, Amazon ECS comes with AWS configuration and operational best practices built-in. It's integrated with both AWS tools, such as Amazon Elastic Container Registry, and third-party tools, such as Docker. This integration makes it easier for teams to focus on building the applications, not the environment. You can

run and scale your container workloads across AWS Regions in the cloud, and on-premises, without the complexity of managing a control plane.

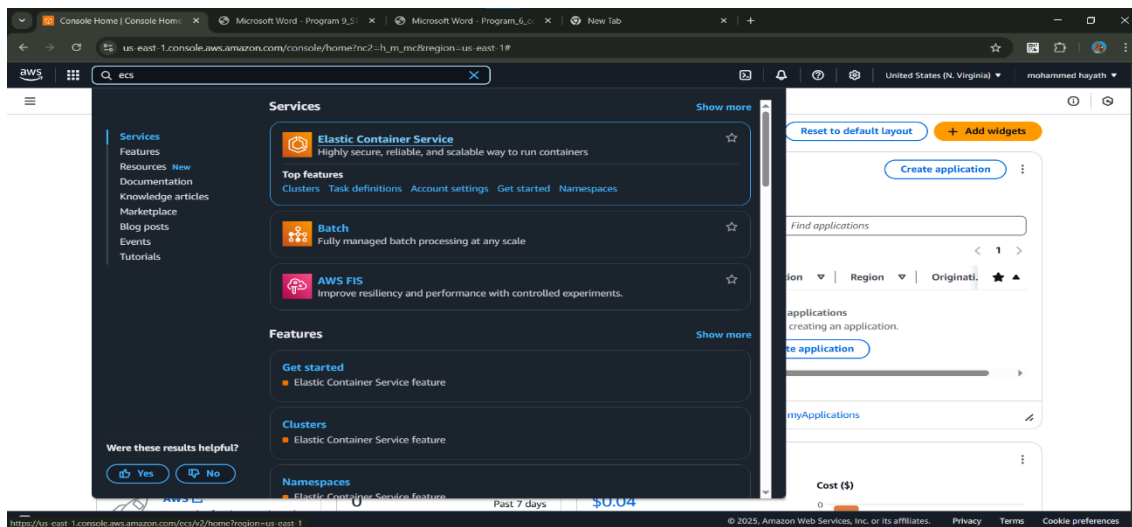
Architect your solution for Amazon ECS

The following diagram shows the application lifecycle and how it works with the Amazon ECS components.

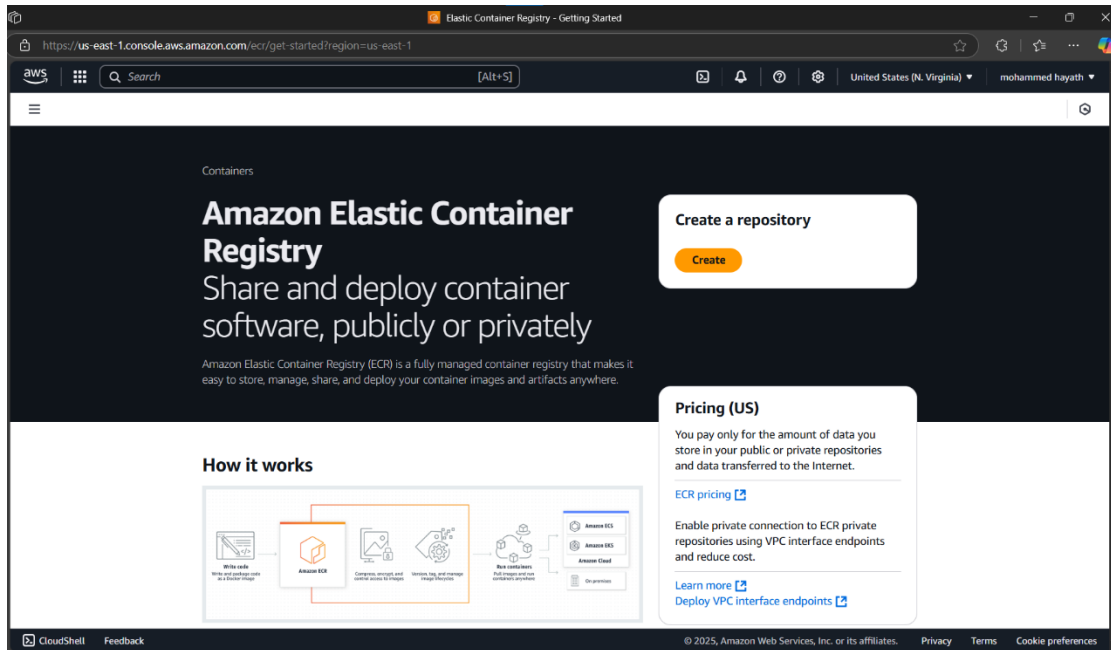
Amazon ECS Application Lifecycle



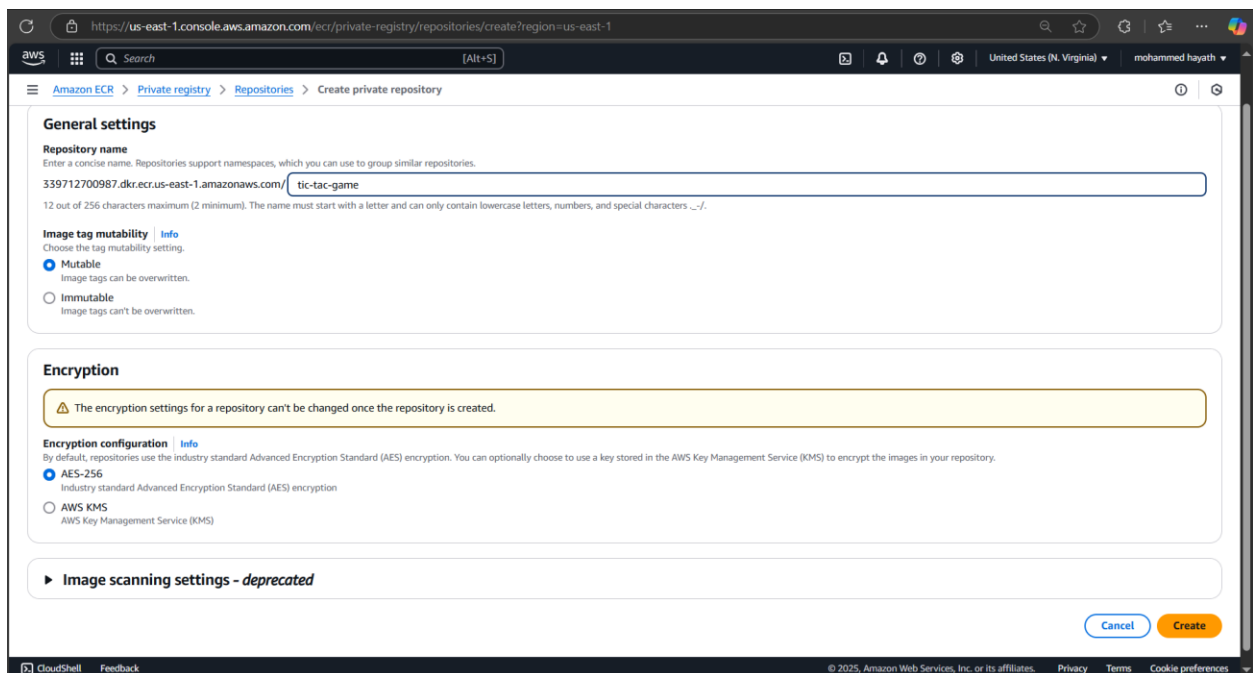
GO TO ➡ Amazon Elastic Container Service ➡ GET STARTED



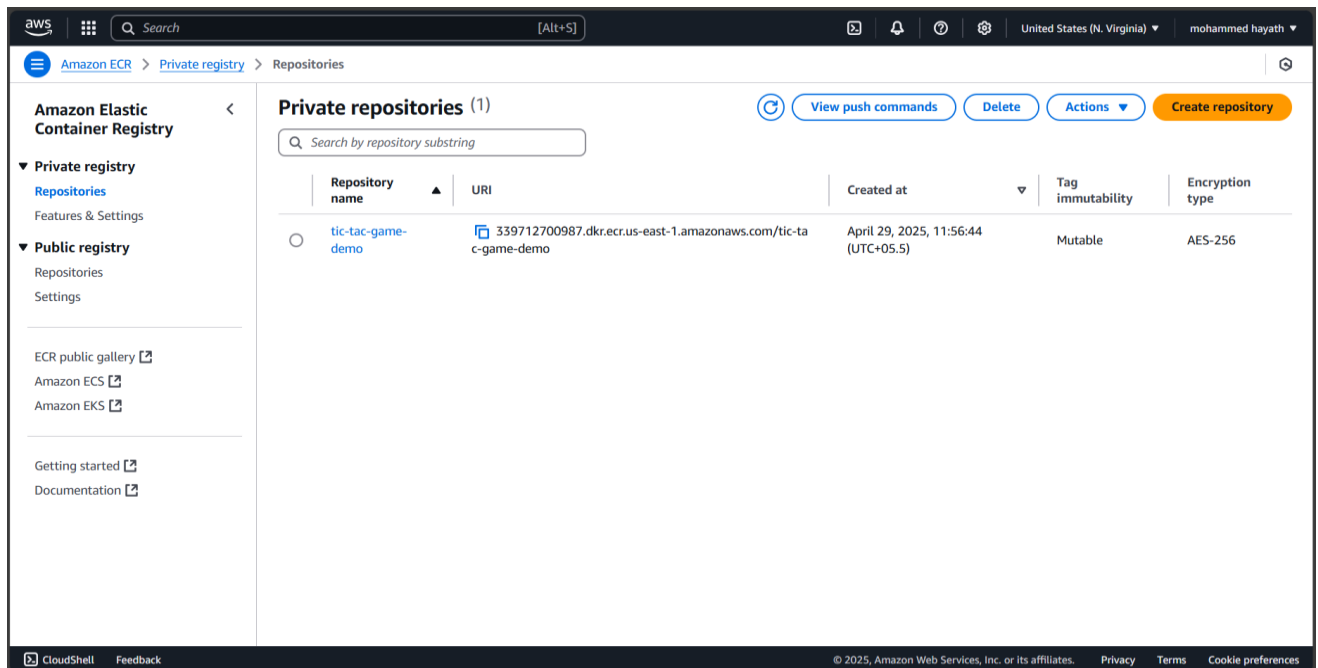
Before creating ECS service create ECR service(Elastic Container Registry)like Github registry which stores container images



Create a repository

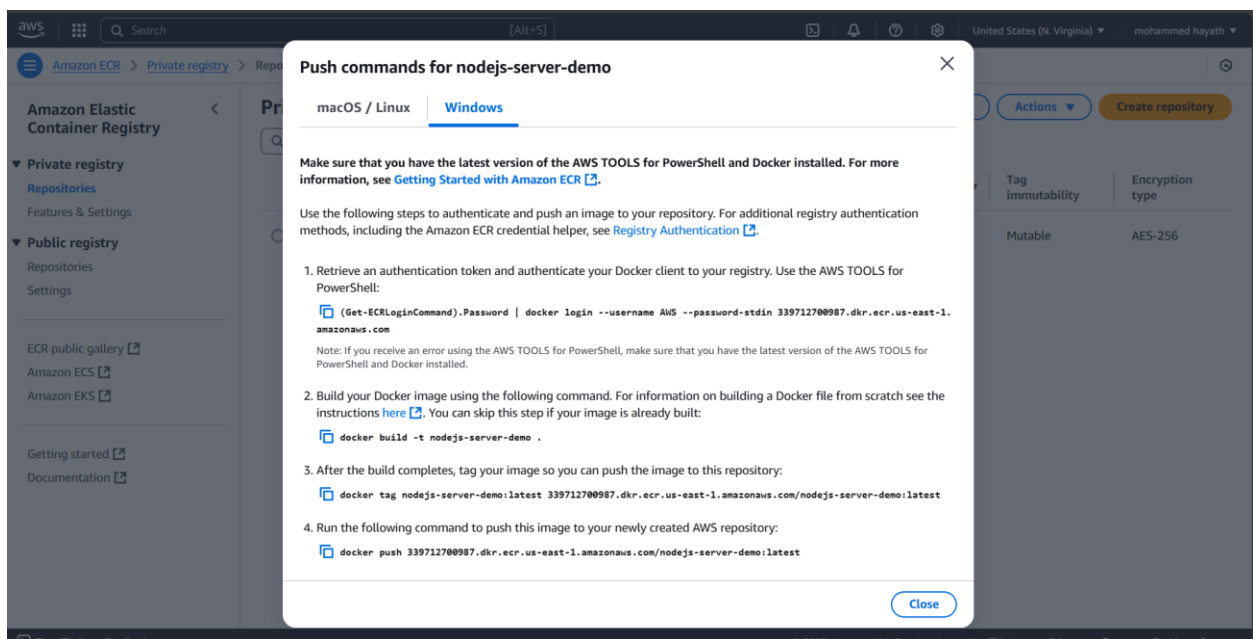


After creating ECR click on push command to push your container images from from your local machine



Click on view push commands

Open terminal in your local machine and navigate to your project directory and run the commands.



Come to your terminal and run the commands to build and run the dockerfile

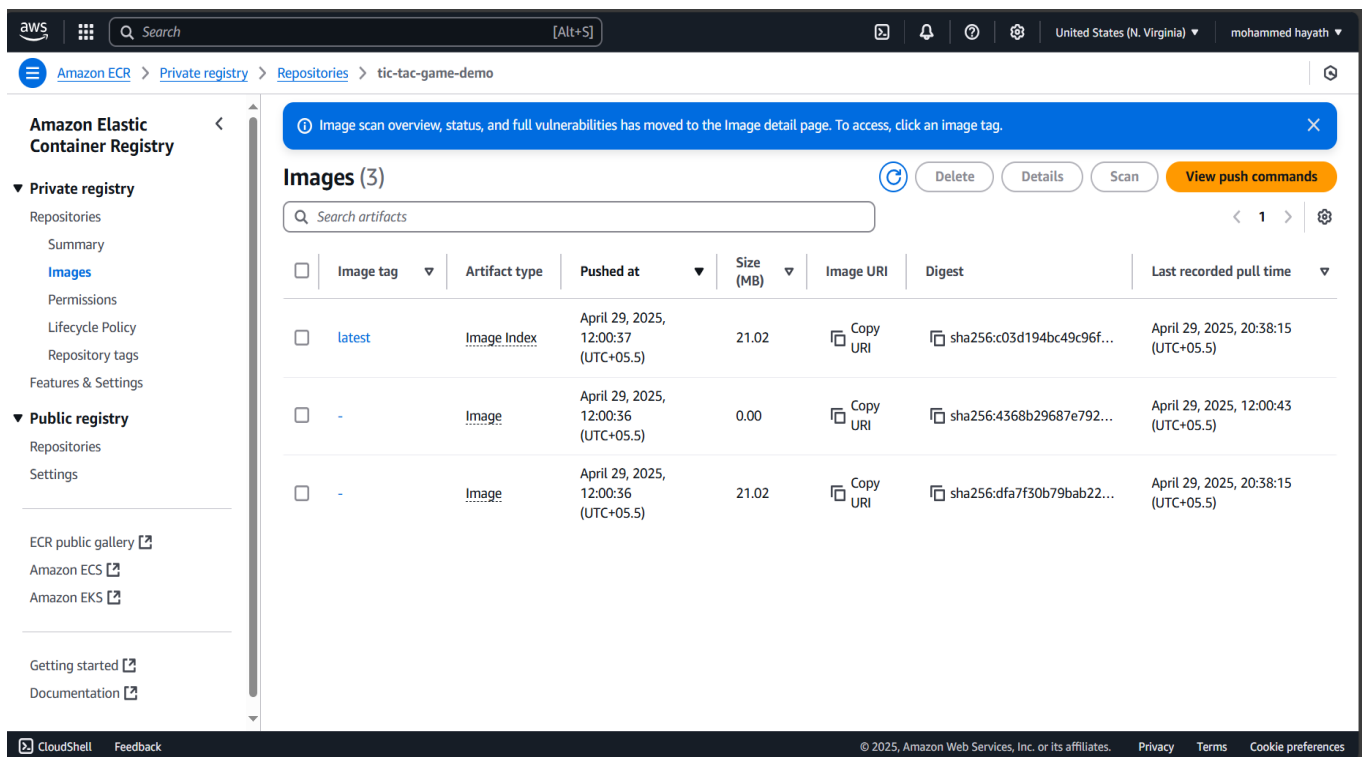
```
Command Prompt
C:\Users\Admin\devsecops-demo>start dockerfile.ext
The system cannot find the file dockerfile.ext.

C:\Users\Admin\devsecops-demo>start Dockerfile.ext
The system cannot find the file Dockerfile.ext.

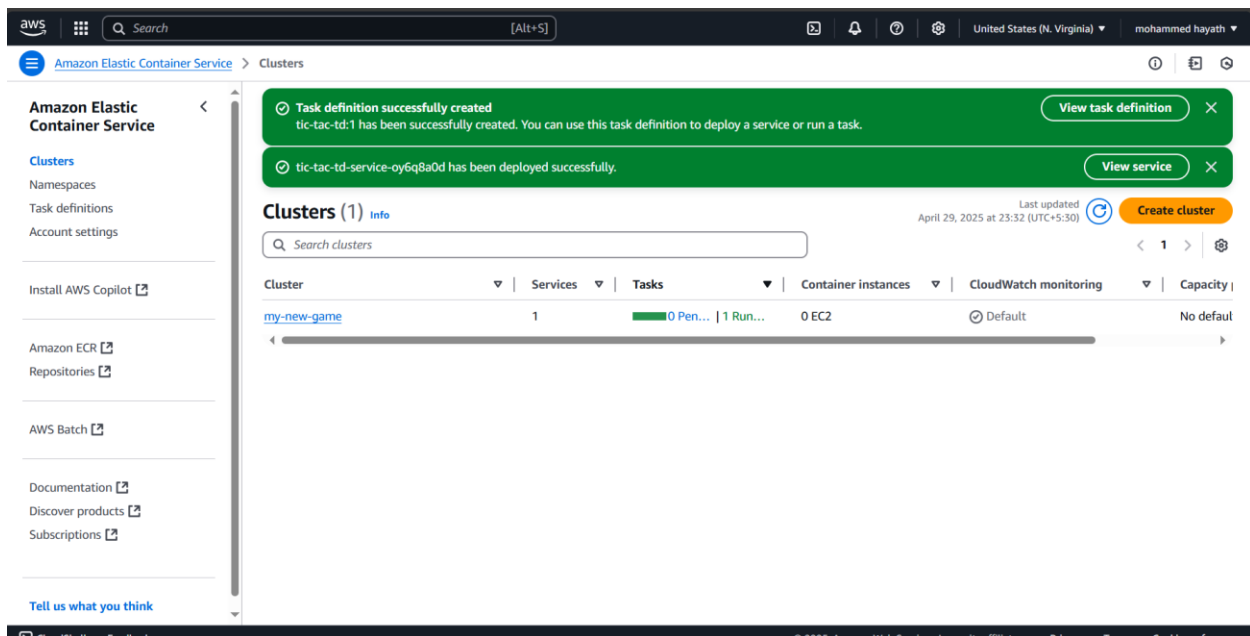
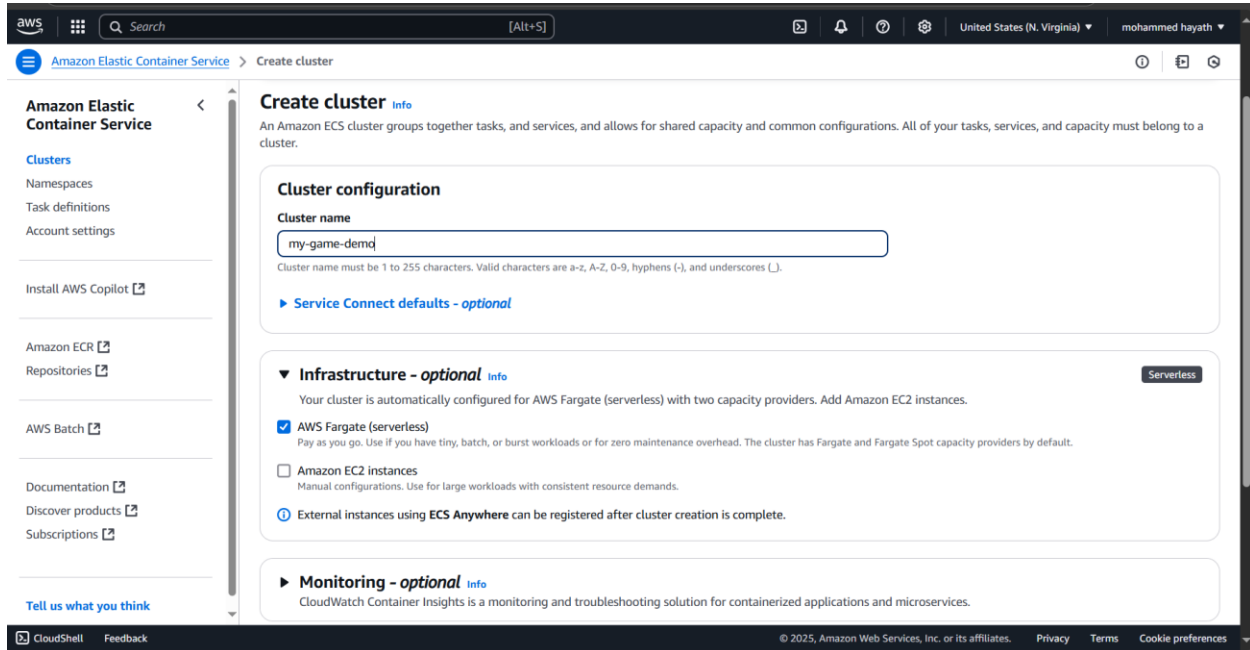
C:\Users\Admin\devsecops-demo>start Dockerfile.txt
The system cannot find the file Dockerfile.txt.

C:\Users\Admin\devsecops-demo>(Get-ECRLoginCommand).Password | docker login --username AWS --password-stdin 339712700987.dkr.ecr.us-east-1.amazonaws.com
```

Open ECR service in aws and you will be able to see the containers which will uploaded from your local machine through cmd



Next come to ECS and create a cluster



Next create a task definition and add container into it which we previously uploaded in ECR and change the security inbound rules and specify the containers ports.

Amazon Elastic Container Service

Clusters

Namespaces

Task definitions

Account settings

Install AWS Copilot

Amazon ECR

Repositories

AWS Batch

Documentation

Discover products

Subscriptions

Tell us what you think

Create new task definition

Container - 1

Container details

Name

dockerfile

Image URI

339712700987.dkr.ecr.us-east-1.amazonaws.com/tic-tac-game-demo

Essential container

Yes

Private registry

Private registry authentication

Port mappings

Container port

80

Protocol

TCP

Port name

web

App protocol

HTTP

Remove

Add port mapping

Read only root file system

Read only

Resource allocation limits - conditional

CPU

7

GPU

1

Memory hard limit

3

Memory soft limit

7

Environment variables - optional

Environment variables

Add individually

Add environment variable

Add from file

Add environment file

CloudShell

Feedback

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7

1

3

7

Environment variables - optional

Environment variables

Add individually

Add environment variable

Add from file

Add environment file

Logging - optional

CPU and memory allocation for a sidecar

We recommend that you use log collection for tasks running on AWS Fargate. Learn more about log collection.

Log collection

Use log collection

Amazon CloudWatch

Key

Value type

Value

awslog-group

Value

/ecs/tic-tac-td

Remove

awslog-region

Value

us-east-1

Remove

awslog-stream-prefix

Value

ecs

Remove

awslog-create-group

Value

true

Remove

mode

Value

non-blocking

Remove

max-buffer-size

Value

25m

Remove

Add log configuration option

Restart policy - optional

HealthCheck - optional

Startup dependency ordering - optional

Container timeouts - optional

Container network settings - optional

Now create a service inside the cluster

The screenshot shows the 'Create service' page in the AWS Management Console for Amazon Elastic Container Service. The left sidebar contains navigation links for Clusters, Namespaces, Task definitions, Account settings, and various AWS services like ECR, Batch, and CloudShell. The main content area is titled 'Create service' and includes the following sections:

- Service details:**
 - Task definition family:** A dropdown menu with 'tic-tac-td' selected.
 - Task definition revision:** A text input field with '1' entered.
 - Service name:** A text input field with 'tic-tac-td-service-lqtnp18d' entered.
- Environment:**
 - Existing cluster:** A dropdown menu with 'my-new-game' selected.
 - Compute configuration (advanced):**
 - Compute options:** A section with two radio buttons: 'Capacity provider strategy' (unselected) and 'Launch type' (selected).
 - Launch type:** A dropdown menu with 'FARGATE' selected.

Next configure vpc and add security group or create new security groups and modify the inbound rule

The screenshot shows the 'Deployment configuration' page in the AWS Management Console for Amazon Elastic Container Service. The left sidebar is the same as the previous screenshot. The main content area is titled 'Deployment configuration' and includes the following sections:

- Service type:** A section with two radio buttons: 'Replica' (selected) and 'Daemon' (unselected).
- Desired tasks:** A text input field with '1' entered.
- Availability Zone rebalancing:** A checkbox labeled 'Turn on Availability Zone rebalancing' which is checked.
- Health check grace period:** A text input field with '0' entered.
- Deployment options:** A section with a link for 'Deployment failure detection'.
- Networking:**
 - VPC:** A dropdown menu with 'vpc-0a2823024191442d5' selected.
 - Subnets:** A section with a dropdown menu showing 'Choose subnets' and a 'Clear current selection' button.

Modify the inbound rules if have more one container add in the below option and add the auto load balancer

Create a new security group

Security group details

Specify the configuration to use when creating the new security group.

Security group name

ecs-gtspulzg

Security group description

Created in ECS Console

Security group name must be 1 to 255 characters. Valid characters are a-z, A-Z, 0-9, underscores (_), hyphens (-), colons (:), forward slashes (/), parentheses (()), hash tags (#), commas (,), at signs (@), brackets ([]), plus signs (+), equal signs (=), ampersands (&), semicolons (;), brackets ({}), exclamation points (!), dollar signs (\$), asterisks (*).

Security group description must be 1 to 255 characters. Valid characters are a-z, A-Z, 0-9, underscores (_), hyphens (-), colons (:), forward slashes (/), parentheses (()), hash tags (#), commas (,), at signs (@), brackets ([]), plus signs (+), equal signs (=), ampersands (&), semicolons (;), brackets ({}), exclamation points (!), dollar signs (\$), asterisks (*).

Inbound rules for security groups

Add one or more ingress rules for your security group.

Type

Custom TCP

Protocol

TCP

Port range

80

Source

Custom

Values

CIDR or IP

Delete

Add rule

Public IP

Choose whether to auto-assign a public IP to the task's elastic network interface (ENI).

Turned on

Service Connect - optional

Service Connect allows for service-to-service communications with automatic discovery using short names and standard ports.

Service discovery - optional

Service discovery uses Amazon Route 53 to create a namespace for your service, which allows it to be discoverable via DNS.

Load balancing - optional

Configure load balancing using Amazon Elastic Load Balancing to distribute traffic evenly across the healthy tasks in your service.

VPC Lattice - optional

Fully managed application networking service to connect, secure, and monitor your services across multiple accounts and virtual private clouds (VPCs). When you use VPC Lattice, there is a cost associated with it.

Service auto scaling - optional

Automatically adjust your service's desired count up and down within a specified range in response to CloudWatch alarms. You can modify your service auto scaling configuration at any time to meet the needs of your application.

Volume - optional

Configure a data volume to provide additional storage for the containers in the task.

Tags - optional

Tags help you to identify and organize your resources.

Cancel>Create

Cluster Service created

Amazon Elastic Container Service

Clusters > my-new-game > Services

Task definition successfully created
tic-tac-tot1 has been successfully created. You can use this task definition to deploy a service or run a task.

tic-tac-tot1 has been deployed successfully.

my-new-game
Last updated April 29, 2025 at 23:51 (UTC+5:30)

Cluster overview

ARN
arn:aws:ecs:us-east-1:339712700987:cluster/my-new-game

Status
Active

CloudWatch monitoring
Default

Registered container instances
-

Services

Draining
Active
1

Tasks

Pending
Running
1

Services (1)

Filter launch type
Any launch type

Filter service type
Any service type

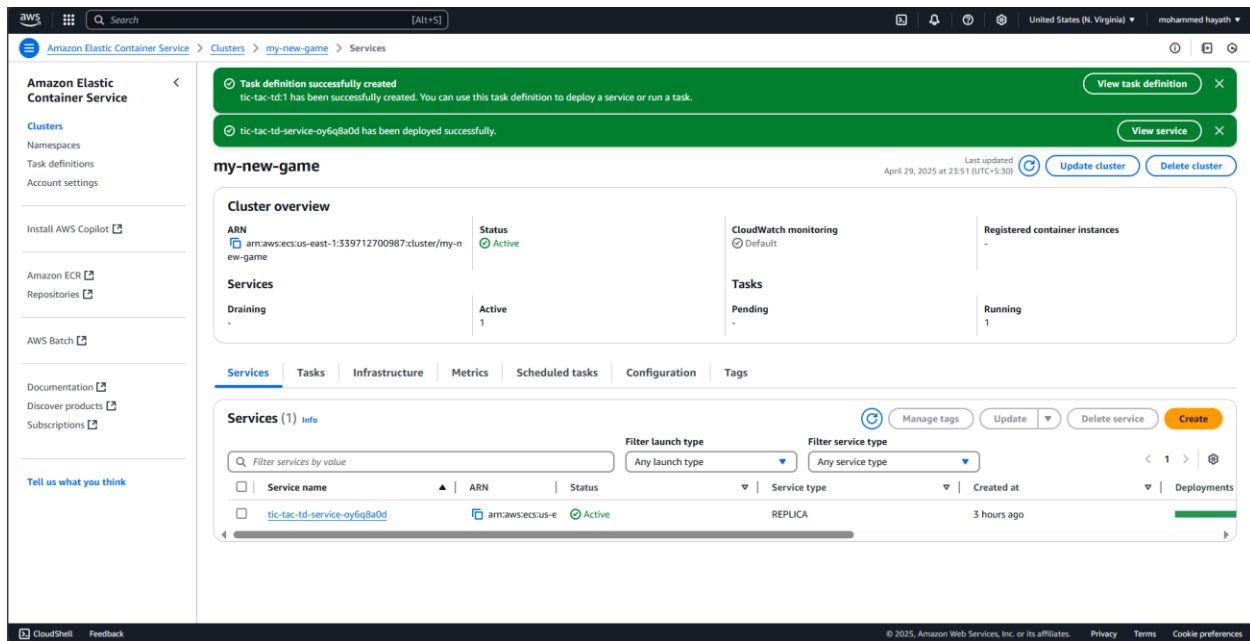
Manage tags
Update
Delete service
Create

Filter services by value

Service name
ARN
Status
Service type
Created at
Deployments

tic-tac-tot-service-oy6q8a0d
arn:aws:ecs:us-east-1:339712700987:cluster/my-new-game
Active
REPLICA
3 hours ago

Copy the DNS from ther service and open load balancer and you will get DNS Url



And paste DNS into your browser

